4.3 Dividing Polynomials

Name _____ Period ____ Date _____

Algebra II 4.3 (30-minute plan) · Student sheet

I can…

- use long division to divide polynomials.
- use synthetic division to divide polynomials by x − k.
- use the Remainder Theorem to evaluate polynomial functions.

Vocabulary · fill in the missing word

Polynomial long division — _____ a polynomial by another polynomial, the same way as long division of numbers

Synthetic division — a shortcut for dividing a polynomial by x − k using only the _____

Remainder Theorem — when f(x) is _____ by x − k, the remainder is f(k)

Part 1 · Notes and examples (fill in each note, then work the example)

Long division: Long division steps: _____, multiply, _____, bring down.

Example 1

Divide $(2x^3 + 3x^2 - x + 5) \div (x^2 - x + 1)$.

Synthetic division: For synthetic division by x − k, put _____ in the box; use _____ for missing terms.

Remainder Theorem: Remainder Theorem: the remainder of f(x) ÷ (x − k) is _____.

Example 2

Divide $(x^3 - 3x^2 - 7x + 6) \div (x - 4)$.

Part 2 · Partner practice

Partner practice

Divide $(x^2 + 7x - 2) \div (x - 1)$.